

A three-dimensional micro-instrumented neural network device

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Three-dimensional (3D) cultured neural networks that emulate the structures and computational principles of the brain could be of use in the development of brain-inspired computing and artificial intelligence, as well as in the understanding of neural development and disease progression. However, creating such stable device–neural network interfaces remains challenging, limiting the potential of such 3D neural networks. Here we report a 3D micro-instrumented neural network device in which a 3D flexible electronic sensor and stimulator array is integrated with a 3D cultured neural network. Our device can be used to record action potentials from multiple planes over a period of 6 months, allowing the quantitative monitoring of the evolving connectivity maps and the pharmacological stimulation responses of the neural networks. This approach also supports chronic electrical stimulation, which we use to train neural networks by tuning the connectivity strengths between neurons, creating a reservoir neural network for biocomputing.

The recent development of artificial intelligence has been largely driven by artificial neural networks¹. However, training artificial neural networks is data-expensive and energy-consuming. Such networks can, in particular, require six-orders-of-magnitude more energy than it takes to perform similar computations in mammalian brains². Thus, understanding the brain's network structures and working principles could help in the development of general-purpose computing with improved data and energy efficiencies^{2,3}, as well as provide insights into the brain's physiology and pathology^{4–7}.

Traditional two-dimensional (2D) neural cultures, though informative^{8,9}, often fail to capture the topology and physiology of in vivo neural connectivity and activity^{10,11}. Their cell–cell and cell–matrix interactions are atypical, with unrealistic gene expressions due to flattened cells, and they generally have low survivability (Supplementary Fig. 1a,b). Advances in tissue engineering have allowed three-dimensional (3D) in vitro neural network models that closely mimic the brain's cytoarchitecture, topology and function to be created, including brain organoids^{4,5} and 3D neural cultures^{12,13}.

Brain organoids—self-assembled 3D aggregates of induced pluripotent stem cells—are a powerful model system for clinical neurological research^{14,15}. However, a lack of precise controllability in cell-type

specification and connectivity formation can lead to inconsistent structure and physiology between samples, resulting in poor replicability and limited application in mechanistic and neuromorphic studies^{16,17}. Brain organoids also suffer from low generation throughput, high heterogeneity, hypoxia and necrosis³. On the other hand, 3D neural cultures with fully controlled cell types, stereotypic cell connections and engineered environments can provide a reproducible in vitro model system that recapitulates the brain's network structures^{10,18} (Supplementary Fig. 1c–f) and could provide a robust platform to create biological neural networks (BNNs) (Fig. 1a). The analogue, localized and distributed information processing of such BNNs—in which dense local neural connections and sparse distant connections (small-world clusters)¹⁹ prevent extensive data movement—could, in particular, provide a route to improved large-scale neuromorphic systems²⁰.

However, the development of methodologies that can provide real-time volumetric monitoring and manipulation of 3D neural cultures over extended periods of time remains challenging. Optical imaging using calcium sensors²¹ or voltage indicators²² can provide the spatial mapping of network dynamics but is constrained by a relatively low volumetric imaging speed and shallow monitoring depth due to light scattering²³. Planar multielectrode arrays (MEAs) can offer multiplexed

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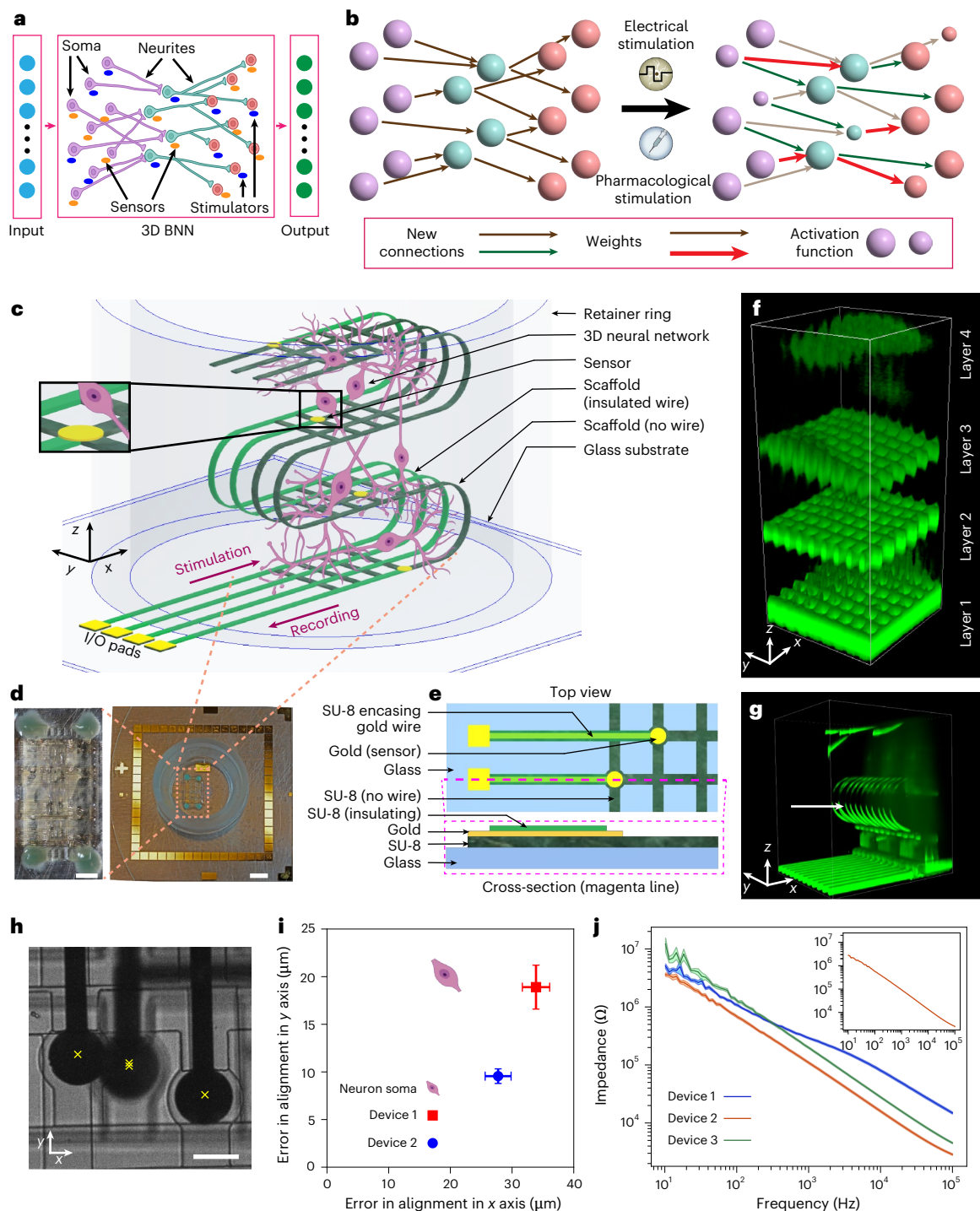


Fig. 1 | Fabrication of 3D-MIND. **a**, Framework of a trackable BNN using a 3D-MIND. **b**, Schematic of an evolving BNN that can be mapped and modulated using a 3D-MIND. **c**, Overview of the device (only 4 out of the 59 sensors are shown). Inset: zoomed-in view of a sensor and a neuron. **d**, Photographs of the fabricated device on the right and zoomed-in view of the 3D-MIND on the left. **e**, Top view shows the schematic of the sensors embedded within flexible polymers (SU-8) on top of a fused silica substrate. Dark green, bottom SU-8; yellow, gold sensors, I/O pads and interconnecting wires; light green, top insulating SU-8; light blue, fused silica substrate. The cross-sectional view shows layers of fused silica, bottom SU-8, sensor, I/O pad, interconnecting wire and insulating top SU-8. **f**, 3D fluorescence image of the folded scaffold. **g**, 3D fluorescence image of the

electrical recordings only from the bottom surface of 3D neural cultures or brain organoids^{3,24}. Flexible MEAs that interface with 3D neural cultures or brain organoids via shell electronics²⁵, mechanical buckling²⁶

or kirigami²⁷ can provide long-term stable electrophysiology. However, the distributions of sensors are still limited to the 2D curved surfaces of 3D neural cultures or brain organoids, preventing the monitoring and

modulation of neuronal dynamics and connectivity inside 3D neural networks. Flexible MEAs can be placed inside organoids by using cell migrations during organogenesis²⁸. However, this leads to an uncontrolled distribution of the electrodes' position and is generally not applicable to 3D neural cultures in which cell migrations happen infrequently. Embedded 3D electrodes on rigid vertical pillars can probe neural activities inside neural networks²⁹. However, such technologies suffer from long-term interface instability because of the mechanical mismatch between the rigid vertical pillars and soft neural networks^{29–31}.

In this Article, we report a 3D micro-instrumented neural network device (3D-MIND) in which an array of electronic sensors and stimulators on a flexible polymer substrate is embedded inside a 3D cultured neural network. The 3D distribution of electronic sensors and stimulators within the 3D neural network can be precisely controlled, allowing volumetric access to targeted neurons. We show that our device can be used to track extracellular action potentials (APs) for over 6 months, detect pharmacological stimulations volumetrically over time and train 3D neural networks by modulating the connectivity strengths between neurons via electrical stimulations over time (Fig. 1b).

Fabrication, assembly and characterization of 3D-MIND

The 3D-MIND was fabricated with close-to-100% yield using standard photolithography similar to procedures described in previous reports^{32,33} (Fig. 1c–g and Supplementary Fig. 2). First, a 2- μm -thick electrically insulating polymer SU-8 was spin coated over a fused silica substrate, which was first selectively patterned with a 100-nm-thick sacrificial nickel layer. Second, SU-8 was photolithographically patterned into a tissue-like mesh structure, followed by a second step of photolithography and metal deposition that defines an array of gold sensors (30 μm diameter) along with their corresponding input–output (I/O) pads (2.2 mm $\times$ 2.2 mm) and interconnecting metal wires (10 μm wide) (Fig. 1e and Supplementary Fig. 2). Third, a second round of SU-8 spin coating and photolithography was performed to selectively insulate the interconnecting metal wires and leave the sensors and I/O pads exposed. Last, an additional SU-8 layer with controlled thickness, ranging from 25 μm to 250 μm , was spin coated and patterned. This SU-8 layer served as the spacer that defines the interlayer spacing and total vertical coverage of the microfabricated device once it was assembled into 3D structures, as described below.

Inspired by origami, we physically folded our device to assemble it from a 2D form after microfabrication into a 3D structure with precisely distributed electronic sensors^{33,34}. First, the nickel layer beneath the SU-8 polymer was etched so that part of the sensor region of the device becomes free-standing (Supplementary Fig. 3a). This free-standing region was then manually folded and glued into multiple stacked layers separated vertically by the spacers, with each layer spanning 5 mm $\times$ 6.5 mm (Supplementary Fig. 3b–f). Importantly, the SU-8 spacers prevent the layers from sagging by creating a controllable gap between each folded layer. After folding, the devices were assembled into 3D tissue-like scaffolds, spanning a total of ~100 μm to ~1 mm in the vertical dimension that was controlled by the spacers (Fig. 1f,g and Supplementary Fig. 4). In addition,

the spacers allowed the alignment and positioning of the sensors across layers with precision comparable with the size of a single neuron soma (Fig. 1h,i and Supplementary Fig. 5). After folding, we characterized the electrical impedance and noise level of the sensors across a frequency range of 10 Hz to 100 kHz (Fig. 1j and Supplementary Fig. 6a). Particularly relevant to AP recording is the performance at 1 kHz, we obtained impedance values of ~200 k Ω (Supplementary Fig. 6b) and noise level of ~80 dB (Supplementary Fig. 6a).

Stable long-term recordings and pharmacological responses

We integrated the assembled 3D devices with 3D cultured neural networks (Methods). First, we dissociated rat hippocampal tissues and cultured the extracted neurons on the substrate of 3D-MIND for 20 min before excess media were aspirated. Second, we created a uniform solution of Matrigel (5 mg ml⁻¹) and neuronal cell suspension (~750k cells ml⁻¹), covering the 3D assembled device. The proteins in the cured Matrigel act as a support for the neurons to form connections and facilitate an environment rich in glial cells³⁵. The presence of glial cells, resembling the brain's *in vivo* microenvironment, is crucial for the long-term growth of the 3D neural networks³⁶. To characterize the interface between our devices and the 3D cultured neural networks, we dyed the SU-8 polymer with rhodamine B and the neurons with CellBrite Green and DAPI. Volumetric confocal fluorescence images highlighted three important features. First, neurons make connections in all 3D directions with extensive neurite growth (Fig. 2a and Supplementary Fig. 1c–f). Second, our 3D-MIND with macroporous structures and multiple folded layers can interpenetrate with the 3D cultured neural networks (Fig. 2b,c and Supplementary Video 1). Third, neurites prefer growing along the SU-8 scaffold: ~75% of tracked axons followed the SU-8 scaffold lines on a plane that constituted ~50% SU-8 surface, allowing for the patterning and positioning of neurons (Fig. 2d,e). Importantly, our 3D-MIND did not disrupt the native neural networks, as confirmed by the unchanged neuronal density with or without the embedded device (Supplementary Fig. 7).

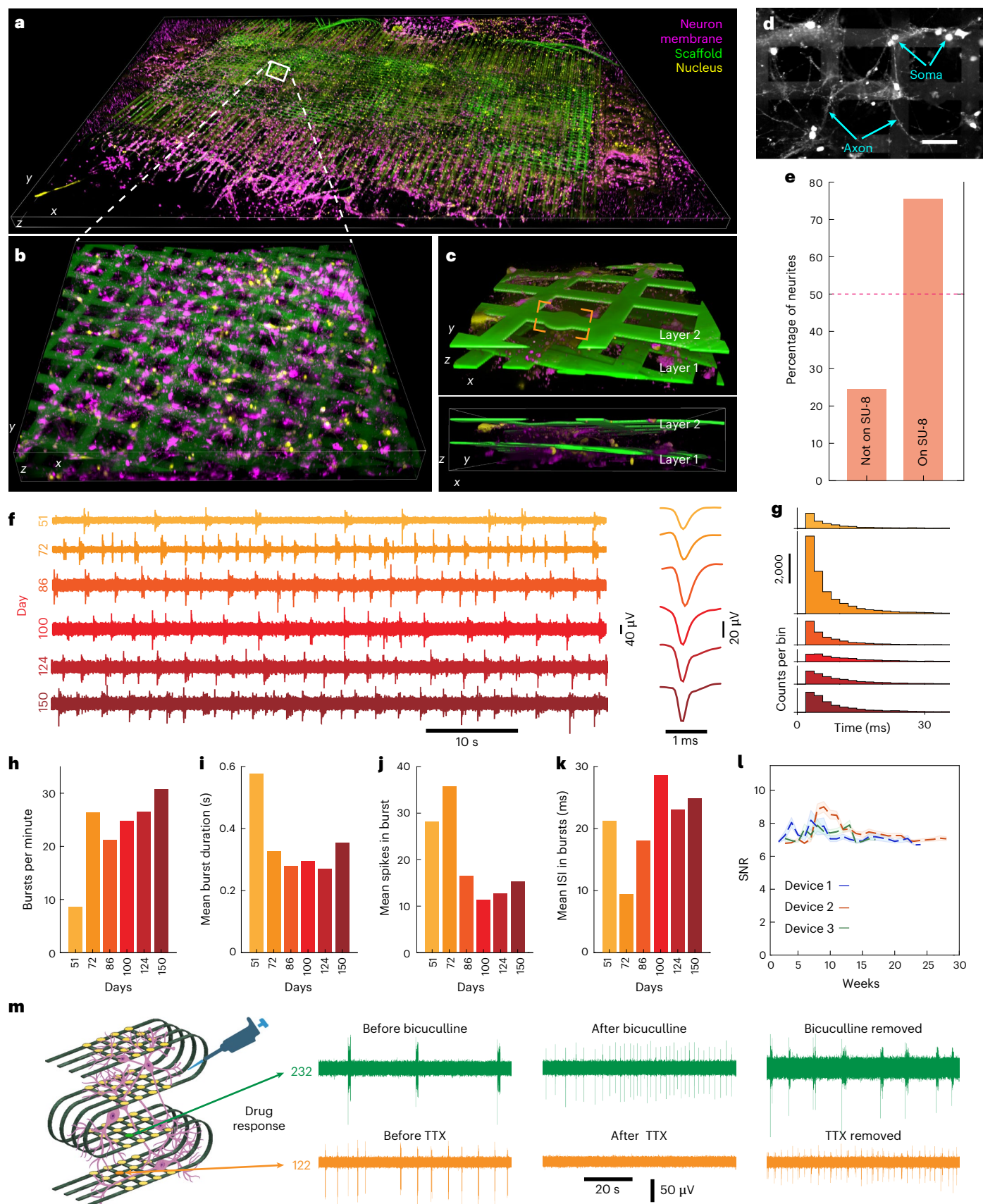
With the 3D interpenetrated interface between our 3D-MIND and the 3D cultured neural networks, we demonstrated the stable long-term tracking of electrical activities from the neural cultures. Electrical recordings from 3D-MIND demonstrated that extracellular APs can be recorded for up to 191 days *in vitro* (DIV) with no decay in the signal amplitude (Fig. 2f,g). This long-term electrical interface stability at the single-neuron level laid down the foundation for our subsequent BNN training via chronic electrical stimulation. Moreover, we observed robust burst activities in our chronic recordings and tracked the evolution of key burst characteristics over time. The frequency and interspike interval of the burst gradually increased as the culture matured over months, whereas the duration of the bursts and number of spikes within each burst decreased (Fig. 2h–k). Importantly, the burst characteristics observed in 3D-MIND differ from those in 2D cultures grown on control planar MEAs. Specifically, in contrast to 2D neural cultures, in which burst activities are relatively frequent with a unimodal distribution, bursts in 3D neural networks occur less frequently with bimodal

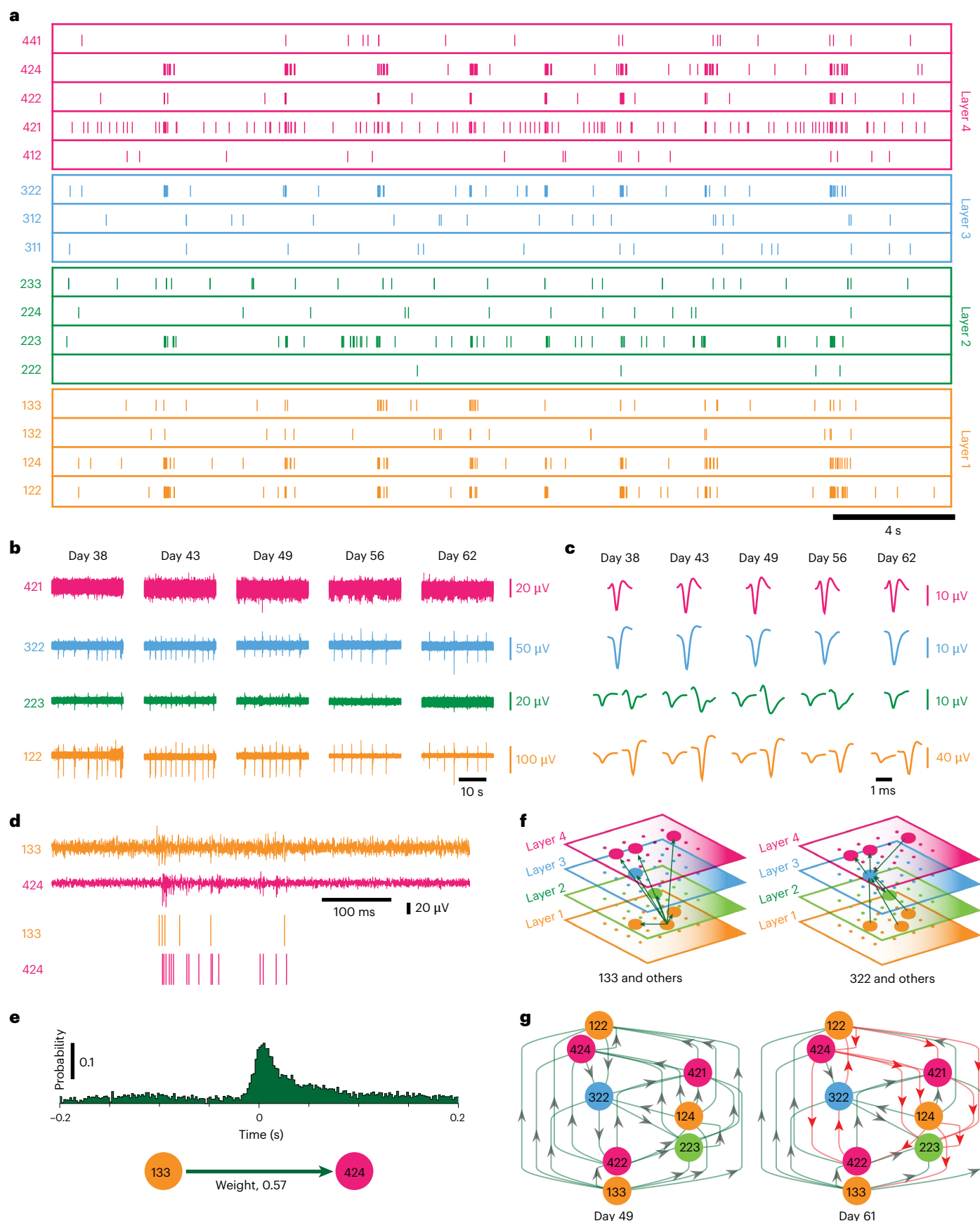
Fig. 2 | Integration with 3D neural network, stable long-term AP recordings and responses to pharmacological stimulations. **a**, Volumetric confocal fluorescence image of 3D-MIND (44 DIV). Magenta, plasma membrane (CellBrite Green); yellow, nucleus (DAPI); green, device (rhodamine B). **b**, Zoomed-in view of the white dashed box in **a**. **c**, 3D (top) and side (bottom) views of two of the layers of the 3D assembled device interfaced with the 3D neural culture; the sensor region is highlighted by the orange dashed box. **d**, A layer of the SU-8 scaffold showing neurites (in white) preferentially growing along the scaffold. **e**, Percentage of neurites growing along the SU-8 scaffold ($n = 15$ neurons). The scaffold plane is 50% porous (red dashed line). **f**, Representative recording traces from day 51 to day 150. Averaged sorted single-unit APs of corresponding traces are shown in the right column. **g**, Interspike interval distributions of

traces shown in **f**. **h–k**, Analyses of burst activities over 6 months: the evolution of burst frequency (**h**), the change in burst duration (**i**), the average number of spikes contained in each burst (**j**) and the mean interspike interval (ISI) within the bursts (**k**). **l**, Averaged signal-to-noise-ratio (SNR) of the sensors over the lifetime of culture ($n = 3$ devices; 10, 11 and 11 sensors from each). The shaded areas indicate ± 1 s.e.m. **m**, Long-term monitoring of pharmacological modulations with 3D-MIND. Results from two drugs, namely, bicuculline (top row) and TTX (bottom row), are shown. The following bounding boxes are used: 7,800 $\mu\text{m} \times 5,300 \mu\text{m} \times 155 \mu\text{m}$ (**a**), 750 $\mu\text{m} \times 750 \mu\text{m} \times 80 \mu\text{m}$ (**b**) and 270 $\mu\text{m} \times 270 \mu\text{m} \times 60 \mu\text{m}$ (**c**). Scale bar, 50 μm (**d**). Illustration in **m** created in BioRender; Mritunjay, K. <https://BioRender.com/x9ueoe> (2026).

distributions (Supplementary Fig. 8a,b). These suggest two distinct types of burst in three dimensions: shorter bursts with fewer spikes that decrease in strength rapidly, and longer bursts that have a slower fall in spike amplitude³⁷. Additionally, compared with neurons in 2D cultures,

neurons in 3D networks exhibit more non-burst activities that resemble the neuronal activities inside the brain (Supplementary Fig. 8c)^{13,31,37–39}. Crucially, the sensors showed stable signal-to-noise ratio over the lifetime of the 3D cultures (Fig. 2l). Together, these results suggest





that 3D neural cultures better recapitulate the fundamental energy and computational principles of the brain²⁰.

To test 3D-MIND as a platform for the 3D chronic monitoring of pharmacological modulations and to confirm that our recorded signals

were physiologically relevant, we investigated 3D-MIND's responses to various drugs (Fig. 2m and Supplementary Fig. 9). We recorded the pre-drug firing activity from our devices, which acted as the baseline, before replacing the culture media with a drug and re-recording the

Fig. 3 | Volumetric electrophysiology mapping from 3D neural networks over months. **a**, Spike trains from 16 representative sensors distributed in four different vertical layers at 49 DIV. Signals are colour coded as follows: orange, layer 1 (bottom); green, layer 2; blue, layer 3; magenta, layer 4 (top). **b**, Representative recording traces across four different vertical layers—one representative channel from each layer—from day 38 to day 62 (same colour code as in **a**). **c**, Averaged sorted APs from the corresponding sensors and days from **b**. **d**, Example recording traces (top) and spike trains (bottom) obtained

simultaneously from sensor 133 in layer 1 (orange) and sensor 424 in layer 4 (magenta) on day 49. **e**, Cross-correlogram of the two channels in **d**, indicating that the neuron recorded by sensor 424 is downstream of the neuron recorded by sensor 133 with a weight of 0.57 (bin, 2 ms). **f**, 3D connectivity mapping of all the channels in relation to sensor 133 (left) and sensor 322 (right). **g**, Inferred network connectivity between different neurons based on the 3D connectivity mapping on day 49 (left) and day 61 (right). The red arrows show the connections that changed over time.

firing activity. The drug was then washed twice with drug-free culture media, and the firing activity was recorded again. We studied 3D-MIND's responses to three drugs: bicuculline—a γ -aminobutyric acid type A receptor antagonist⁴⁰, tetrodotoxin (TTX)—a sodium channel blocker⁴¹ and cyanquixaline (CNQX)—an α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid-type glutamate receptor antagonist⁴². First, γ -aminobutyric acid type A receptors are ligand-gated ion channels that have an inhibitory influence on target neurons⁴³; therefore, a γ -aminobutyric acid type A receptor antagonist like bicuculline increases the likelihood of a neuron firing. As expected, the addition of bicuculline in 3D-MIND increased the firing rate of the neurons by almost sevenfold in a few minutes, and the effect persisted after the bicuculline was washed. The firing rate went back to its pre-drug rate 24 h after the bicuculline was removed (Fig. 2m (top row) and Supplementary Fig. 9a). The increase in the firing rate is synonymous with lowering the threshold of the activation function of the nodes, thereby allowing us to pharmacologically modulate the training of the BNN. Second, opposite to bicuculline, the addition of TTX immediately suppressed the firing activities of all neurons (Fig. 2m (bottom) and Supplementary Fig. 9b), suggesting a rise in the threshold of the activation function. To ensure that the disappearance of neuronal firing activity was not caused by neuronal death, we removed the TTX 48 h after the initial TTX addition. Immediately after the TTX was removed, the firing activities of neurons recovered with firing rates returning to pre-TTX level (Fig. 2m (bottom) and Supplementary Fig. 9b). Last, the addition of CNQX temporarily silenced burst firings in the neurons. The burst firings recovered after CNQX was removed (Supplementary Fig. 9c).

Volumetric electrophysiology mapping from 3D neural networks

Next, we investigated the 3D long-term mapping capability of 3D-MIND during the development of the 3D cultured neural networks. For the convenience of description, we number the sensors in an x - y - z format where x represents the layer (1 denotes the bottom layer and 4, the topmost layer), and y and z represent the sensor row and column within the layer, respectively. First, electrical recordings from 16 sensors distributed into 4 layers, with 3–5 sensors selected from each layer, at 49 DIV show clear single-unit activities from three dimensions, confirming the volumetric mapping capabilities of 3D-MIND (Fig. 3a). Importantly, this 3D volumetric electrophysiology mapping is stable across all four layers over the long term, as evidenced by the selected (one from each layer) sensors' recording traces and their corresponding

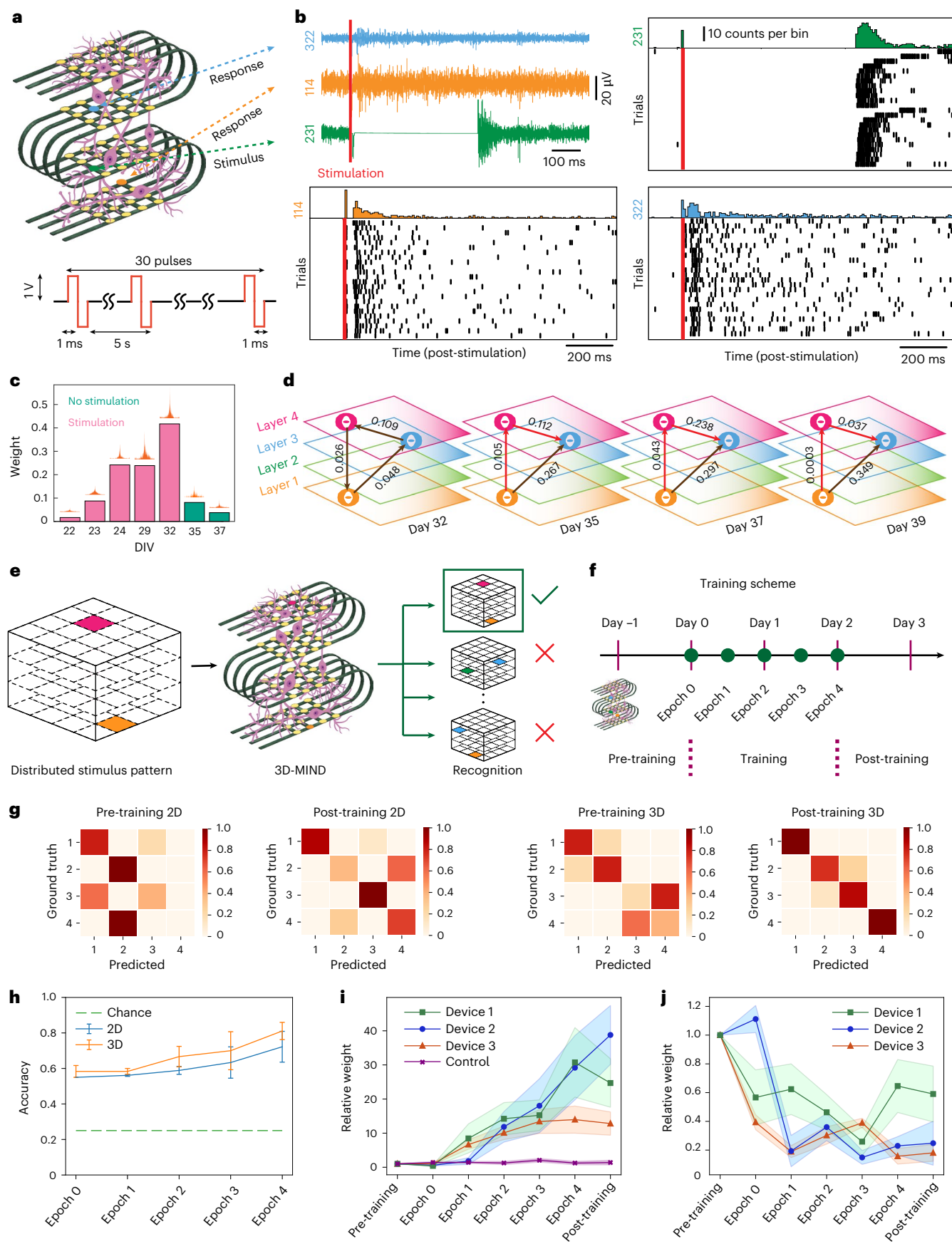
sorted spikes (Fig. 3b,c). Second, the 3D volumetric electrophysiology mapping capability of 3D-MIND allowed us to track the neuronal connectivity as the 3D cultured neural networks evolved. Specifically, we used a cross-correlogram constructed from the spike trains to infer the functional connectivity and information flow direction between pairs of neurons^{44,45} (Fig. 3d). To quantitatively estimate the neuronal connectivity, we performed Gaussian fitting on the resulting cross-correlogram and used the metric obtained from the Gaussian fitting as an indicator of the connectivity strength/weight (Methods and Supplementary Fig. 10a). The cross-correlogram of sensor 424 with respect to sensor 133 shows that the neuron recorded by 424 is downstream of the neuron recorded by 133 (Fig. 3e) with a weight of 0.57. Last, we generalized this analysis pipeline to all the sensors to construct the 3D connectivity map of the 3D cultured neural networks (Fig. 3f and Supplementary Fig. 10b). We tracked the 3D connectivity map over time and found that the connectivity map evolved over the course of time, as is expected in vivo (Fig. 3g).

Modulating network connectivity and constructing BNNs

We further expanded the capabilities of 3D-MIND by incorporating electrical stimulations. As a proof of concept, we first tested whether chronic electrical stimulations could modulate the 3D neural connectivity by altering the neural connections and/or their weights. We chronically stimulated neurons from a given layer and observed neuronal responses across the 3D volume (Fig. 4a and Supplementary Fig. 12a,b). As an example, we applied electrical pulse trains with 1-ms pulse duration and 0.2-Hz repetition rate (Fig. 4a, bottom) to stimulator 231 (layer 2) and recorded the activities from all sensors. Representative recording traces from sensor 231 (layer 2), sensor 114 (layer 1) and sensor 322 (layer 3) were analysed (Fig. 4b). A perievent raster plot of these sensors show that the electrical stimulation of neurons near sensor 231 can lead to an increased firing activity in neurons recorded by sensors 114 and 322, suggesting functional connections between these neurons (Fig. 4b and Supplementary Fig. 12c–f). Applying the cross-correlogram analyses described above, we tracked the connectivity strength between neurons recorded by sensors 322 and 421 over multiple days while applying electrical stimulation via stimulator 231. Interestingly, we found that the connectivity between these neurons strengthened over time, suggesting long-term plasticity⁴⁶ akin to evolving weights between nodes in artificial neural networks (Fig. 4c). The increased trend, however, fades away after the electrical stimulation was withdrawn, suggesting a dynamic

Fig. 4 | Modulating neuronal connectivity via chronic electrical stimulations and constructing BNNs. **a**, Schematic showing the concept of spatially patterned stimulation and recording of 3D-MIND (top) and the stimulation parameters (bottom). **b**, Representative trace and perievent raster plot of three sensors, namely, 322 (layer 3, blue), 114 (layer 1, orange) and 231 (layer 2, green), in response to electrical stimuli applied to sensor 231 (the red line represents stimulation time). Each black vertical line in the perievent raster represents a spike event from the corresponding sensor ($n = 24$ stimulation trials). Corresponding histograms of spike distributions is shown above the perievent raster. There is a blank space in the 231 trace because of stimulation artifacts. **c**, Evolution of the weight between neurons recorded by sensors 322 and 421 over multiple days in response to electrical stimulation applied to sensor 231. **d**, Neurons adjacent to sensors 114, 322 and 421 changed their network properties in response to stimulation, going

from a feedback loop to a directional network. **e**, Schematic showing the use of 3D-MIND as a reservoir neural network. **f**, Training scheme of the neural network. **g**, Representative confusion matrices before and after training using 2D and 3D distributed sensors. **h**, Changes in prediction accuracy with training using 2D and 3D distributed sensors. The green line represents the accuracy obtained by chance. Data are presented as mean values $\pm$ s.e.m. **i**, j, Evolving connectivity strength in response to stimulations. The weights were normalized with respect to the corresponding prestimulation weights of each pair. **i**, Connection getting stronger in response to stimulation ($n = 3$ neural cultures, 10 pairs of neurons from each culture). **j**, Connection getting weaker in response to stimulation ($n = 3$ neural cultures, 4 pairs of neurons from each culture). The shaded areas indicate ± 1 s.e.m. Illustrations in **a**, **e** and **f** created in BioRender; Mritunjay, K. <https://BioRender.com/x9ueoe> (2026).



plasticity in neural connectivity⁴⁷. Plasticity observed in 3D-MIND as a result of 0.2-Hz stimulation is consistent with previous reports^{48–53}. We then used the volumetric recording capability of 3D-MIND to track the evolving neural network in response to external stimulations. As a demonstration, we analysed the connectivity between three neurons near sensors 114, 322 and 421 across days in response to chronic stimulation. Interestingly, we observed the net directionality between 322 and 421, and between 114 and 421, reversed along with changes in their weights (Fig. 4d). Utilizing the plasticity of neuronal connectivity and the long-term stability of our 3D device–3D neural network interface, we used 3D-MIND to construct a reservoir neural network for classification. First, we input different spatially patterned biphasic electrical pulses (Fig. 4a, bottom) from the volumetrically distributed stimulators of 3D-MIND (Fig. 4e). Next, we utilized the neuronal responses recorded from the sensors to fit a logistic regression model to classify the different input patterns (Fig. 4e, Supplementary Figs. 13 and 14 and Methods). We trained the neural networks over 2 days with an epoch every 12 h (Fig. 4f). In addition, to test our hypothesis that 3D device interfaces outperform 2D interfaces, we trained, in parallel, a logistic regression model with the outputs only from sensors sitting at the bottom layer of the same cultures. We found that first, representative confusion matrices for both 2D and 3D models showed improved classification accuracy after training (Fig. 4g). In specific, the naive 3D-MIND started from a relatively low accuracy of classification at $58.3 \pm 3.3\%$ (mean $\pm$ 1 s.e.m), which improved to $81.1 \pm 4.8\%$ after four epochs of training (Fig. 4h). Second, we observed that the 3D device interfaces outperformed their 2D counterparts at each intermediate epoch, with faster learning and higher classification accuracy at the end of the four epochs (Fig. 4h). The learning of 3D-MIND can be attributed to unsupervised learning through changes in network connectivity over the course of training³. Last, to quantify the evolution of 3D neural networks in response to training, we tracked the connectivity strength among neural populations and observed a substantial increase in the weight between some pairs of neurons (Fig. 4i) and a substantial decrease in others (Fig. 4j).

Conclusions

We have reported a 3D flexible bioelectronics device with precisely positioned sensors and stimulators that can be used to form 3D device–3D neural network architectures. Our device can be embedded inside and can interpenetrate 3D cultured neural networks without perturbing the morphology or distribution of the neurons. It can be used to track the activity, connectivity and development of the 3D cultured neural networks for over 6 months. In addition, 3D-MIND can be used to monitor the responses of the 3D cultured neural networks to pharmacological stimulations, as well as to train the 3D cultured neural networks by modulating the connectivity between neurons with spatially patterned electrical stimulations over days. Furthermore, we showed that 3D-MIND can function as a tunable and fully observable reservoir neural network.

3D-MIND has a number of advantages compared with existing methodologies. The soft scaffold structure can form non-invasive interpenetration within a 3D neural network and preserve its cyto-architecture and computational characteristics, allowing long-term usage. The 3D locations of the embedded sensors and stimulators can be precisely distributed both laterally and axially, allowing controlled volumetric access to neurons within the cultured neural network. The interfaced system can then provide a physiologically relevant understanding of the brain's 3D network connectivity.

With the ability to track a 3D neural network, 3D-MIND could be of use in understanding the efficiency and versatility of the brain's computational capabilities, and can be applied in the development of general-purpose brain-inspired computing. Although we mainly used a stimulation frequency of 0.2 Hz, the device is capable of other stimulation parameters. Comparing connectivity changes in response to different stimulation frequencies may help control the training rates and

reveal potential learning rules and mechanisms. 3D-MIND could also be further improved by increasing the number of sensors⁵⁴, types of device⁵⁵ and number of folded planes for highly multiplexed multimodal studies. In addition, although single-neuron-level alignment of the sensors and stimulators can be achieved in three dimensions, the manual folding process is not yet scalable. Its reproducibility and throughput may pose challenges for mass production. To reach industry-level production, the automation of the folding process via self-folding²⁵, buckling²⁶ or other innovative approaches with precision comparable with the size of a single neuron soma (tens of micrometres) would be essential.

Methods

3D-MIND fabrication

The key steps in the fabrication process are shown in Supplementary Fig. 2. The scaffold parameters are as follows: width of each layer of scaffold, 6.3 mm; length of each layer of scaffold, 5 mm; scaffold ribbon width, 25 μ m; longitudinal gap between ribbons, 50 μ m; transverse gap between ribbons, 75 μ m; width of metal interconnect lines, 10 μ m; diameter of sensors, 30 μ m. The key steps involved in fabrication are as follows. (1) Lift-off resist LOR3A (Kayaku Advanced Materials) and positive photoresist (PR) AZ 5214 (EMD Performance Materials) were spin coated on a 500- μ m-thick 100-mm-diameter fused silica wafer (University Wafer), which was precleaned using acetone and isopropanol (IPA), dehydrated at 180 °C for 5 min and further cleaned using oxygen plasma. AZ 5214 was patterned and exposed using a laser writer (Heidelberg DWL 66+) and developed in AZ 300 MIF developer (EMD Performance Materials) for 1 min, and a 100-nm-thick sacrificial layer of nickel was deposited (Angstrom Nexdep). Lift-off was done in MR 1165 (DuPont Electronics Materials) at 80 °C for 1 h. (2) A 2- μ m-thick negative PR SU-8 2002 (Kayaku Advanced Materials) was spin coated (bottom SU-8), followed by soft baking at 65 °C and 95 °C for 1 min each. SU-8 was patterned and exposed using a mask aligner (Suss MA6) and then post-exposure baked at 65 °C and 95 °C for 1 min each. The PR was developed using an SU-8 developer solution (Kayaku Advanced Materials) for 1 min, rinsed with IPA and dried with nitrogen, followed by a hard bake at 185 °C for 1 h. (3) LOR3A was spin coated on the sample followed by a layer of positive PR AZ 1505 (EMD Performance Materials). The sample was then selectively exposed using a mask aligner (Suss MA6) and developed to create the pattern for metal sensors, interconnects and I/O pads, followed by the thermal deposition of 10-nm chromium and 100-nm gold using Angstrom Covap. Lift-off was done in MR 1165 at 80 °C for 6 h, and the sample was then rinsed with deionized water and IPA, and was dried with nitrogen. (4) Step 2 was repeated to create another layer of SU-8 to passivate the metal lines, leaving the sensors and I/O pads exposed; hard baking was done at 195 °C. (5) 250- μ m-thick negative PR SU-8 2075 (Kayaku Advanced Materials) was spin coated (Spacer SU-8), followed by soft baking at 65 °C and 95 °C for 10 min and 3.5 h, respectively. The PR was patterned and exposed using a mask aligner (Suss MA6) and then post-exposure baked at 65 °C and 95 °C for 5 min and 16 min, respectively. The PR was developed for 45 min using an SU-8 developer solution, rinsed with IPA and dried with nitrogen, followed by a hard bake at 205 °C for 1 h. (6) The sample was then exposed to an oxygen plasma (100 W, 1 min) to make the SU-8 hydrophilic. (7) The wafer was then coated with another protective layer of PR AZ 5214 and diced in the shape of the final MEA using an ADT Dicing saw.

3D-MIND assembly

After the PR in the fabricated device was removed using acetone and IPA, it was then put into a wet nickel etchant TFB (Transene) to remove the sacrificial nickel layer. The removed nickel allows one end of the scaffold to float, whereas the other end stays attached to the substrate (Supplementary Fig. 3a). The device was then rinsed in deionized water and the scaffold was folded. The key steps in folding are as follows. (1) The device was submerged in deionized water and a pipette was used to create a gentle flow such that the free end folds at 180°, the

folded end was held in place by a thin-tipped pipette as the water was gently removed and a finer alignment of the scaffold was done under a microscope. (2) The water was allowed to evaporate, and the sides of the scaffold were glued using Kwik-Cast (World Precision Instruments) (Supplementary Fig. 3d). The glue was allowed to cure for 15 min. (3) Step 1 was repeated two more times to create a total of four layers (Supplementary Fig. 3b,c). (4) The entire device was then submerged in 70% ethanol for sterilization, after which a custom-made retainer ring made of acrylic was glued (using polydimethylsiloxane–SYLGARD 184) to the centre of the device to create a well for the media (Supplementary Fig. 3e,f). A lid was also designed to cover the well to reduce evaporation and allow the exchange of gases.

3D-MIND characterization

After the retainer ring was glued to the substrate, it was filled with 1× phosphate-buffered saline (PBS) to characterize the sensors. The device was inserted in the MEA2100-Mini-System (Multi Channel Systems) and the baseline noise from each channel was visually noted. Any sensor that had a very high noise profile was considered to be potentially broken. A sinusoidal pulse of 1 kHz and 1-mV peak-to-peak amplitude (Agilent 33250A function generator) was applied to the PBS buffer using a platinum electrode. The response of the sensors to this signal was visually inspected and was used to further classify any faulty sensors. To characterize the noise profile of the sensors, unfiltered noise was recorded using the MEA2100-Mini-System and its Fourier transform was obtained (Supplementary Fig. 6a). Electrochemical impedance spectroscopy was performed using a lock-in amplifier (Zurich Instruments) to measure the impedance of the sensors (1-kHz sinusoidal voltage; 100-mV peak-to-peak amplitude) (Supplementary Fig. 6b). The electrochemical impedance spectroscopy setup was further used to characterize the frequency response of the sensors by doing a frequency sweep from 10 Hz to 100 kHz (Fig. 1j). The stimulation response of the MEA2100-Mini-System was then characterized by measuring the voltage response of the sensor to an injected current or voltage (Supplementary Fig. 11). The Mini-System was used to inject biphasic current pulses with amplitudes of 5 µA, 10 µA, 20 µA, 50 µA and 100 µA, each with pulse durations of 20 µs, 40 µs, 60 µs, 80 µs, 100 µs, 200 µs, 400 µs and 1 ms. The voltage response of the stimulation in the PBS solution was recorded by an oscilloscope (Tektronix MDO4034-3) using a platinum electrode. Similarly, the response of the sensors to voltage stimulation was also measured by applying biphasic voltage pulses of amplitudes 100 mV, 200 mV, 500 mV, 800 mV and 1 V, each with pulse durations of 20 µs, 40 µs, 60 µs, 80 µs, 100 µs, 200 µs, 400 µs and 1 ms. After sensor characterizations, PBS was removed, and the device was washed twice using cell-grade water (Intermountain Life Sciences) and finally sterilized with 70% ethanol.

Pre-culture treatment of 3D-MIND and control planar MEAs

Poly-D-lysine (Sigma-Aldrich) was diluted in a borate buffer (pH 8.3) to get a 500 µg ml⁻¹ solution. Then, 1.5 ml of this poly-D-lysine solution was added to the culture wells of 3D-MIND and control planar MEAs that were precleaned based on the company's recommendations. The devices were stored in the incubator for 24 h, after which the solution was aspirated, and the wells were washed twice with cell-grade water (Intermountain Life Sciences). Then, 1.5 ml of a 10 µg ml⁻¹ solution of laminin (Corning) in Hanks's balanced salt solution (HBSS; Ca and Mg free; HyClone) was added to the culture wells, and they were stored in the incubator for another 24 h. The laminin was then aspirated, and the wells were washed twice with cell-grade water before proceeding with cell culture.

3D and 2D neural cultures

Seventeen-day-pregnant Sprague Dawley rats (Charles River) were euthanized using CO₂, the embryos were extracted, and the head was removed and stored in HBSS (Cytiva) in an ice bath^{56,57}. Dissection was performed under a

HEPA filter hood during which the embryo heads were stored in a Petri dish over ice. The hippocampus was dissected from the heads in an HBSS-filled Petri dish and the hippocampal tissues were stored in Hibernate E (Gibco) media. To culture the neurons, the hippocampal tissues were washed twice with HBSS in a centrifuge (30 s, 86g). After the second wash, 900 µl of HBSS and 100 µl of 1X trypsin (MP Biomedicals) were added to the tissues, and it was kept in a hot-water bath (37 °C) for 15 min, gently shaking the tube halfway through. The trypsin was washed with HBSS twice in a centrifuge, and after the final wash, 2 ml of neuronal media (Neurobasal media + B-27 (2%) + PBS (1%) + NGF 2.5S (0.1%); all from Gibco) was added to the tissues. A flame-polished Pasteur pipette was used to break the tissues into individual cells. The cell density was calculated using a Hemacytometer (Hausser Scientific) in a Trypan Blue dye (Sigma-Aldrich). The cell solution was appropriately diluted to obtain a cell density of ~700 cells µl⁻¹ and a portion of this cell solution was separated for the 2D seeding of cells. Then, 50 µl of the ~700 cells µl⁻¹ cell solution was added to the base of 3D-MIND and control MEAs to create a 2D cell layer, the devices were then kept still inside the hood for 20 min to allow for the neurons to adhere to the surface. The remaining cell solution was increased to a volume of 5 ml and then centrifuged, the media were aspirated and fresh media were added based on the first cell hemacytometer reading to create a high-density cell solution. During this process, Matrigel (Corning) was thawed in an ice bath. Depending on the density of Matrigel supplied by the manufacturer, the high-density cell solution was added to the Matrigel and thoroughly mixed, resulting in a final protein concentration of ~5 mg ml⁻¹ and a cell density of ~750 cells µl⁻¹. After allowing 20 min for the 2D cells to adhere, the excess media were removed from the substrate. Fresh pre-warmed media were added to the control MEAs and they were moved to the incubator (37 °C). The Matrigel solution was then added to the folded scaffold and the device was moved to the incubator (37 °C) for 45 min for the Matrigel to cure. After 45 min, 1.5 ml of neuronal media were added to the device. The device was then kept in a humidified CO₂ (5%) incubator at 37 °C. Half of the media were first changed after 24 h, and subsequently, half of the media were changed every 2 days.

Electrophysiology recording and stimulation

We used MEA2100-Mini-System to record the electrical activity (at 20 kHz) from the culture placed inside the incubator with Multi Channel Experimenter. We applied a second-order Butterworth bandpass filter (200 Hz to 3,500 Hz) to the raw data. The cultures were recorded for 100 s (a small file for ease of visualization) and 15 min (long recording for data collection and analysis). Different electrical stimulation parameters were tried^{58,59} (Methods section '3D-MIND characterization' and Supplementary Fig. 11) to measure the response of 3D-MIND. A biphasic 1-V peak-to-peak pulse with a duration of 1 ms was chosen for stimulation as it gave the best response. The stimulation pulse was administered every 5 s (0.2 Hz) for a total of 30 pulses (Fig. 4a).

Data processing

NeuroExplorer 5.436 (Nex Technologies) was used to analyse the data. The filtered data generated from our recording setup were used to extract spikes using a threshold of $4 \times$ (median of the filtered signal). All other data processing, such as spike sorting, burst analyses (Max-Interval method⁶⁰) and cross-correlogram (bin information in figures), was also performed in NeuroExplorer. Second-order Gaussian fit of cross-correlograms was done in MATLAB R2024a, resulting in a curve of the form $a1 \times \exp(-\frac{(x-b1)^2}{2c1^2}) + a2 \times \exp(-\frac{(x-b2)^2}{2c2^2})$. A ratio of the sum of the height of the curves and the sum of the standard deviation of the curves $(a1 + a2)/(c1 + c2)$ was used as the parameter to quantify the weight of the connection (connection strength).

Reservoir neural network

For classification-based neural network training (Supplementary Fig. 13), four stimulators (A, B, C and D) were chosen from different layers of 3D-MIND to inject the stimulation pulses. Two out of these

four stimulators were simultaneously stimulated and acted as a single input, for a total of four different input patterns (AB, BC, CD and DA). This allowed an overlap between the different inputs, showcasing the classification property of 3D-MIND for inputs with substantial common traits. A 4.5-min-long recording session was performed that consisted of 30 stimulation pulses, which followed and preceded 1 min of no stimulation. The evoked neural response from 3D-MIND was recorded and converted into a one-dimensional array X to be fed into a logistical regression model. Specifically, the signal traces from $t = 0$ to $t = 300$ ms after each stimulation pulse from the most active sensors were converted into a spike train, which were then binned into 2-ms time windows to form an array X_i . Such arrays (X_i) from each sensor were then appended to each other (retaining the same order in all training steps) to form the final array $X = [X_1 X_2 \dots X_n]$ (Supplementary Fig. 14). Here 120 such arrays X (30 stimulation pulses from each of the 4 inputs) acted as the dataset for training a logistic regression model for classification. The training of classification weights was done by the scikit-learn package in Python 3.12.7, with 75% samples used for training the model and the remaining used for testing the accuracy. The stimulation input patterns were given to three different neural cultures with a set of epochs every 12 h for a total of five epochs. The training accuracies were obtained after each epoch.

Pharmacological stimulation

Here 10 mM of bicuculline in dimethylsulfoxide (HY-N0219) was obtained from MedChemExpress. Then, 15 μ l of this solution was mixed with 1,485 μ l of pre-warmed culture media to make a working bicuculline solution with a concentration of 100 μ M. The cell culture was removed from the incubator and the culture media were aspirated. Then, 1,500 μ l of the working bicuculline solution was added to the cell culture. Electrophysiology was done keeping the culture in the incubator. The bicuculline solution was then washed twice (5 min each time) using fresh pre-warmed media. TTX was obtained from Cayman Chemicals (14963). Then, 1 mg of TTX was diluted in 3.134 ml of citrate buffer solution (Sigma-Aldrich; C2488-100ML) to make a 1-mM TTX solution. Thereafter, 1 ml of the resulting solution was further diluted in 9 ml of sterile water to make a 100- μ M solution; 15 μ l of this solution was added to 1,485 μ l of pre-warmed culture media to make the final TTX solution with a concentration of 1 μ M. The cell culture was removed from the incubator and the media were aspirated. The TTX solution was added to the culture and electrophysiology was performed. The TTX was washed twice (5 min each) with fresh pre-warmed media after 48 h. CNQX was obtained from Cayman Chemicals (14618). Then, 1.6 mg of CNQX was diluted in 1.380 ml of dimethylsulfoxide (Thermo Scientific, J66650.AK) to make a 5-mM CNQX solution; 50 μ l of this solution was added to 450 μ l of pre-warmed culture media to make the final working CNQX solution with a concentration of 500 μ M. The cell culture was removed from the incubator and 100 μ l of the working CNQX solution was added to the culture, and electrophysiology was performed. The CNQX was washed twice (5 min each) with fresh pre-warmed media.

Fluorescence imaging

Cultures were grown and interfaced with scaffold structures on glass-bottomed 12-well plate (Mattek) for fluorescence imaging. A dye solution was made using 1.5 ml of fresh pre-warmed media, 15 μ l of 1 mg ml⁻¹ DAPI (Thermo Fisher) and 8 μ l of CellBrite Green (Biotium). Culture media were replaced by the dye solution in the neural culture, and it was incubated at 37 °C for 1 h. The dye solution was washed with fresh pre-warmed media and stored in the incubator at 37 °C for 1 h; the washing process was repeated one more time. The media were then washed using Dulbecco's phosphate-buffered saline (DPBS) and incubated for 5 min; this wash step was also done twice. The neural culture was fixed by replacing DPBS with 4% paraformaldehyde in PBS (Thermo Scientific) and leaving it at room temperature for 15 min. Paraformaldehyde was then washed with DPBS at room temperature for 3 min and this step was repeated two more times. The fixed neural culture

was imaged in a DPBS solution using a Nikon Ti2 inverted microscope with Yokogawa W1 and Sora modules. The images were captured and processed using NIS-Elements AR 5.42.02 (Build 1801).

Ethical compliance

This work was reviewed and approved by Princeton University's Institutional Animal Care and Use Committee (IACUC). The animals were housed in an AAALAC International accredited facility that adheres to principles stated in the Guide for the Care and Use of Laboratory Animals (8th Edition).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

References

1. LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444 (2015).
2. Mehonic, A. & Kenyon, A. J. Brain-inspired computing needs a master plan. *Nature* **604**, 255–260 (2022).
3. Cai, H. et al. Brain organoid reservoir computing for artificial intelligence. *Nat. Electron.* **6**, 1032–1039 (2023).
4. Paşca, S. P. The rise of three-dimensional human brain cultures. *Nature* **553**, 437–445 (2018).
5. Chiaradia, I. & Lancaster, M. A. Brain organoids for the study of human neurobiology at the interface of in vitro and in vivo. *Nat. Neurosci.* **23**, 1496–1508 (2020).
6. Kang, Y. J. et al. Three-dimensional human neural culture on a chip recapitulating neuroinflammation and neurodegeneration. *Nat. Protoc.* **18**, 2838–2867 (2023).
7. Paşca, S. P. et al. A nomenclature consensus for nervous system organoids and assembloids. *Nature* **609**, 907–910 (2022).
8. Isomura, T., Kotani, K. & Jimbo, Y. Cultured cortical neurons can perform blind source separation according to the free-energy principle. *PLoS Comput. Biol.* **11**, e1004643 (2015).
9. Sumi, T. et al. Biological neurons act as generalization filters in reservoir computing. *Proc. Natl Acad. Sci. USA* **120**, e2217008120 (2023).
10. Jensen, C. & Teng, Y. Is it time to start transitioning from 2D to 3D cell culture? *Front. Mol. Biosci.* **7**, 33 (2020).
11. Kagan, B. J. et al. The technology, opportunities, and challenges of Synthetic Biological Intelligence. *Biotechnol. Adv.* **68**, 108233 (2023).
12. Anderson, W. A., Bosak, A., Hogberg, H. T., Hartung, T. & Moore, M. J. Advances in 3D neuronal microphysiological systems: towards a functional nervous system on a chip. *In Vitro Cell. Dev. Biol. Anim.* **57**, 191–206 (2021).
13. Frega, M., Tedesco, M., Massobrio, P., Pesce, M. & Martinoia, S. Network dynamics of 3D engineered neuronal cultures: a new experimental model for in-vitro electrophysiology. *Sci. Rep.* **4**, 5489 (2014).
14. Lancaster, M. A. et al. Cerebral organoids model human brain development and microcephaly. *Nature* **501**, 373–379 (2013).
15. Paşca, A. M. et al. Functional cortical neurons and astrocytes from human pluripotent stem cells in 3D culture. *Nat. Methods* **12**, 671–678 (2015).
16. Quadrato, G., Brown, J. & Arlotta, P. The promises and challenges of human brain organoids as models of neuropsychiatric disease. *Nat. Med.* **22**, 1220–1228 (2016).
17. Andrews, M. G. & Kriegstein, A. R. Challenges of organoid research. *Annu. Rev. Neurosci.* **45**, 23–39 (2022).

18. Ravi, M., Paramesh, V., Kaviya, S. R., Anuradha, E. & Solomon, F. D. 3D cell culture systems: advantages and applications. *J. Cell. Physiol.* **230**, 16–26 (2015).
19. Bassett, D. S. & Bullmore, E. T. Small-world brain networks revisited. *Neuroscientist* **23**, 499–516 (2017).
20. Vianello, E. & Payvand, M. Scaling neuromorphic systems with 3D technologies. *Nat. Electron.* **7**, 419–421 (2024).
21. Zhang, Y. et al. Fast and sensitive GCaMP calcium indicators for imaging neural populations. *Nature* **615**, 884–891 (2023).
22. Abdelfattah, A. S. et al. Sensitivity optimization of a rhodopsin-based fluorescent voltage indicator. *Neuron* **111**, 1547–1563.e9 (2023).
23. Xu, C., Nedergaard, M., Fowell, D. J., Friedl, P. & Ji, N. Multiphoton fluorescence microscopy for in vivo imaging. *Cell* **187**, 4458–4487 (2024).
24. Obien, M. E. J., Deligkaris, K., Bullmann, T., Bakkum, D. J. & Frey, U. Revealing neuronal function through microelectrode array recordings. *Front. Neurosci.* **8**, 423 (2015).
25. Huang, Q. et al. Shell microelectrode arrays (MEAs) for brain organoids. *Sci. Adv.* **8**, eabq5031 (2022).
26. Park, Y. et al. Three-dimensional, multifunctional neural interfaces for cortical spheroids and engineered assembloids. *Sci. Adv.* **7**, eabf9153 (2021).
27. Yang, X. et al. Kirigami electronics for long-term electrophysiological recording of human neural organoids and assembloids. *Nat. Biotechnol.* **42**, 1836–1843 (2024).
28. Le Floch, P. et al. Stretchable mesh nanoelectronics for 3D single-cell chronic electrophysiology from developing brain organoids. *Adv. Mater.* **34**, 2106829 (2022).
29. Soscia, D. A. et al. A flexible 3-dimensional microelectrode array for in vitro brain models. *Lab Chip* **20**, 901–911 (2020).
30. Spanu, A. et al. A three-dimensional micro-electrode array for in-vitro neuronal interfacing. *J. Neural Eng.* **17**, 036033 (2020).
31. Shin, H. et al. 3D high-density microelectrode array with optical stimulation and drug delivery for investigating neural circuit dynamics. *Nat. Commun.* **12**, 492 (2021).
32. Fu, T.-M. et al. Stable long-term chronic brain mapping at the single-neuron level. *Nat. Methods* **13**, 875–882 (2016).
33. Dai, X., Zhou, W., Gao, T., Liu, J. & Lieber, C. M. Three-dimensional mapping and regulation of action potential propagation in nanoelectronics-innervated tissues. *Nat. Nanotechnol.* **11**, 776–782 (2016).
34. Sturm, J. C., Giles, M. D. & Gibbons, J. F. A three-dimensional folded dynamic RAM in beam-recrystallized polysilicon. *IEEE Electron Device Lett.* **5**, 151–153 (1984).
35. Kleinman, H. K. & Martin, G. R. Matrigel: basement membrane matrix with biological activity. *Semin. Cancer Biol.* **15**, 378–386 (2005).
36. Araque, A. & Navarrete, M. Glial cells in neuronal network function. *Philos. Trans. R. Soc. B* **365**, 2375–2381 (2010).
37. Cerina, M., Piastra, M. C. & Frega, M. The potential of in vitro neuronal networks cultured on micro electrode arrays for biomedical research. *Prog. Biomed. Eng.* **5**, 032002 (2023).
38. Bourke, J. L. et al. Three-dimensional neural cultures produce networks that mimic native brain activity. *J. Tissue Eng. Regen. Med.* **12**, 490–493 (2018).
39. Cabrera-Garcia, D. et al. Early prediction of developing spontaneous activity in cultured neuronal networks. *Sci. Rep.* **11**, 20407 (2021).
40. Johnston, G. A. Advantages of an antagonist: bicuculline and other GABA antagonists. *Br. J. Pharmacol.* **169**, 328–336 (2013).
41. Kao, C. Y. Tetrodotoxin, saxitoxin and their significance in the study of excitation phenomena. *Pharmacol. Rev.* **18**, 997–1049 (1966).
42. Blake, J. F., Brown, M. W. & Collingridge, G. L. CNQX blocks acidic amino acid induced depolarizations and synaptic components mediated by non-NMDA receptors in rat hippocampal slices. *Neurosci. Lett.* **89**, 182–186 (1988).
43. Masiulis, S. et al. GABA_A receptor signalling mechanisms revealed by structural pharmacology. *Nature* **565**, 454–459 (2019).
44. Aertsen, A. M. H. J. & Gerstein, G. L. Evaluation of neuronal connectivity: sensitivity of cross-correlation. *Brain Res.* **340**, 341–354 (1985).
45. Kobayashi, R. et al. Reconstructing neuronal circuitry from parallel spike trains. *Nat. Commun.* **10**, 4468 (2019).
46. Van Pelt, J., Vajda, I., Wolters, P. S., Corner, M. A. & Ramakers, G. J. A. Dynamics and plasticity in developing neuronal networks in vitro. *Prog. Brain Res.* **147**, 171–188 (2005).
47. Miyamoto, E. Molecular mechanism of neuronal plasticity: induction and maintenance of long-term potentiation in the hippocampus. *J. Pharmacol. Sci.* **100**, 433–442 (2006).
48. Bologna, L. L. et al. Low-frequency stimulation enhances burst activity in cortical cultures during development. *Neuroscience* **165**, 692–704 (2010).
49. le Feber, J., Stegenga, J. & Rutten, W. L. C. The effect of slow electrical stimuli to achieve learning in cultured networks of rat cortical neurons. *PLoS ONE* **5**, e8871 (2010).
50. Vajda, I. et al. Low-frequency stimulation induces stable transitions in stereotypical activity in cortical networks. *Biophys. J.* **94**, 5028–5039 (2008).
51. Ide, A. N., Andruska, A., Boehler, M., Wheeler, B. C. & Brewer, G. J. Chronic network stimulation enhances evoked action potentials. *J. Neural Eng.* **7**, 016008 (2010).
52. Mainen, Z. F. & Sejnowski, T. J. Reliability of spike timing in neocortical neurons. *Science* **268**, 1503–1506 (1995).
53. Shahaf, G. & Marom, S. Learning in networks of cortical neurons. *J. Neurosci.* **21**, 8782–8788 (2001).
54. Vivenzi, J. et al. Flexible, foldable, actively multiplexed, high-density electrode array for mapping brain activity in vivo. *Nat. Neurosci.* **14**, 1599–1605 (2011).
55. Won, S. M., Song, E., Reeder, J. T. & Rogers, J. A. Emerging modalities and implantable technologies for neuromodulation. *Cell* **181**, 115–135 (2020).
56. Kaech, S. & Banker, G. Culturing hippocampal neurons. *Nat. Protoc.* **1**, 2406–2415 (2006).
57. Pacifici, M. & Peruzzi, F. Isolation and culture of rat embryonic neural cells: a quick protocol. *J. Vis. Exp.* **63**, e3965 (2012).
58. Cogan, S. F. Neural stimulation and recording electrodes. *Annu. Rev. Biomed. Eng.* **10**, 275–309 (2008).
59. Boehler, C., Carli, S., Fadiga, L., Stieglitz, T. & Asplund, M. Tutorial: guidelines for standardized performance tests for electrodes intended for neural interfaces and bioelectronics. *Nat. Protoc.* **15**, 3557–3578 (2020).
60. Cotterill, E., Charlesworth, P., Thomas, C. W., Paulsen, O. & Eglén, S. J. A comparison of computational methods for detecting bursts in neuronal spike trains and their application to human stem cell-derived neuronal networks. *J. Neurophysiol.* **116**, 306–321 (2016).

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Author contributions

K.M., T.-M.F. and J.C.S. conceived the project. K.M. and J.C.S. initiated the work on electrical instrumentation and 3D neural cultures. K.M. and T.-M.F. designed the 3D electronics devices and integration with 3D neural cultures. K.M. performed the experiments, analysed the data and wrote the paper. K.M., T.-M.F. and J.C.S. discussed the results and revised the paper.

Competing interests

The authors have filed an international patent application with the USPTO (PCT/US2025/054156) for the device mentioned in this work.

Additional information

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Sample size We tested our system by interfacing 30 3D-MIND devices with neural cultures. The data used in the manuscript are taken from 12 of these cultures.

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Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	17 Day pregnant female Sprague Dawley rats (Charles River) were euthanized to extract embryos.
Wild animals	N/A
Reporting on sex	Embryos were obtained from female pregnant Sprague dawley rats. Hippocampal tissues were extracted from theses embryos. The sex of embryos were not determined as it was not relevant to our work.
Field-collected samples	N/A
Ethics oversight	This work was reviewed and approved by Princeton University's Institutional Animal Care and Use Committee (IACUC). The animals were housed in an AAALAC International accredited facility that adheres to principles stated in the Guide for the Care and Use of Laboratory Animals (8th Edition).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A